Data Cleaning Reasoning Report - Team 5 - Sprint 3

Initial approach: Prepare the Poker Hand History (PHH) dataset by removing irrelevant information, standardizing the data, and optimizing the structure to reduce preprocessing/parsing time and to convert to numpy and pandas objects for scalability. Keep only No-Limit Texas Hold'em data and split it by venue to identify outliers from possible venue card bias. Analyze the flop and hole distributions to calculate deviation from expected values in both independent counts and hands.

Findings:

- Not all of the data has a winnings section, we can "fix" that by replaying the games assuming that the best hand of those that were revealed won.
- Most of the HandHQ portion of the dataset contains hands that were never revealed, giving us less visibility.
- Minimal data to outright remove (see high level inventory section below)

Justifications:

- Justification for trimming down to No-Limit Texas Hold 'em:
    - This is the most common form of poker and the vast majority of the data entries. If we were to include multiple forms of poker we would not be measuring the same types of behavior, and it's important to keep it standard.
- Justification for selecting particular datasets (human vs AI, such as Pluribus)
    - By using only human training data, we ensure that there are no biases introduced by other algorithms.
- Justification for reformatting the dataset
    - Conforming to a traditional database style that has good support for Python libraries like Pandas, Numpy, or Pyarrow will be much more efficient to work with for both training and testing the model.
    - The compute from the Coursera platform VMs appears to run*significantly* slower than even a mid-range laptop, saving time is a must.

Flop data:

Legend

- Left side indicates what vendor (ABS = Absolute Poker)
    - Right side of that row indicates the observed/expected values for the most under/over represented cards (2c = 2 of clubs)
- Rainbow: 3 unique suits (H D S)
- Two-tone: 2 of a suit (H H S)
- Monotone: 3 identical suits (H H H)
- Unpaired: 3 different ranks (8 K Q)
- Paired: 2 of a # (8 8 Q)
- Trips: 3 of a kind (8 8 8)

Image shows actual vs expected (exp). The HandHQ data is nearly as expected from all venues.

```
ABS          2,988,584 cards   per-card obs/exp range 0.9768 (As) .. 1.0202 (2s)
          flop suits: rainbow 39.77% (exp 39.76%)  two-tone 55.08% (exp 55.06%)  monotone 5.15% (exp 5.18%)
          flop ranks: unpaired 82.82% (exp 82.82%)  paired 16.94% (exp 16.94%)  trips 0.24% (exp 0.24%)
FTP          2,450,000 cards   per-card obs/exp range 0.9669 (As) .. 1.0244 (3s)
          flop suits: rainbow 39.71% (exp 39.76%)  two-tone 55.11% (exp 55.06%)  monotone 5.19% (exp 5.18%)
          flop ranks: unpaired 82.76% (exp 82.82%)  paired 17.00% (exp 16.94%)  trips 0.24% (exp 0.24%)
IPN         10,059,688 cards   per-card obs/exp range 0.9648 (Ah) .. 1.0241 (3h)
          flop suits: rainbow 39.76% (exp 39.76%)  two-tone 55.07% (exp 55.06%)  monotone 5.18% (exp 5.18%)
          flop ranks: unpaired 82.77% (exp 82.82%)  paired 16.98% (exp 16.94%)  trips 0.24% (exp 0.24%)
ONG          3,104,075 cards   per-card obs/exp range 0.9659 (Ad) .. 1.0293 (2c)
          flop suits: rainbow 39.70% (exp 39.76%)  two-tone 55.08% (exp 55.06%)  monotone 5.23% (exp 5.18%)
          flop ranks: unpaired 82.88% (exp 82.82%)  paired 16.89% (exp 16.94%)  trips 0.23% (exp 0.24%)
PS          18,360,134 cards   per-card obs/exp range 0.9779 (Ah) .. 1.0170 (2h)
          flop suits: rainbow 39.77% (exp 39.76%)  two-tone 55.03% (exp 55.06%)  monotone 5.20% (exp 5.18%)
          flop ranks: unpaired 82.80% (exp 82.82%)  paired 16.96% (exp 16.94%)  trips 0.24% (exp 0.24%)
PTY          5,790,341 cards   per-card obs/exp range 0.9679 (As) .. 1.0225 (2c)
          flop suits: rainbow 39.64% (exp 39.76%)  two-tone 55.17% (exp 55.06%)  monotone 5.19% (exp 5.18%)
          flop ranks: unpaired 82.77% (exp 82.82%)  paired 16.98% (exp 16.94%)  trips 0.24% (exp 0.24%)
misc                5 cards   per-card obs/exp range 0.0000 (Ac) .. 10.4000 (Jc)
          flop suits: rainbow 0.00% (exp 39.76%)  two-tone 100.00% (exp 55.06%)  monotone 0.00% (exp 5.18%)
          flop ranks: unpaired 100.00% (exp 82.82%)  paired 0.00% (exp 16.94%)  trips 0.00% (exp 0.24%)
pluribus       22,616 cards   per-card obs/exp range 0.8990 (3d) .. 1.1588 (4d)
          flop suits: rainbow 38.97% (exp 39.76%)  two-tone 55.53% (exp 55.06%)  monotone 5.51% (exp 5.18%)
          flop ranks: unpaired 83.31% (exp 82.82%)  paired 16.50% (exp 16.94%)  trips 0.19% (exp 0.24%)
wsop               22 cards   per-card obs/exp range 0.0000 (Ac) .. 4.7273 (As)
          flop suits: rainbow 40.00% (exp 39.76%)  two-tone 60.00% (exp 55.06%)  monotone 0.00% (exp 5.18%)
          flop ranks: unpaired 80.00% (exp 82.82%)  paired 20.00% (exp 16.94%)  trips 0.00% (exp 0.24%)
HandHQ      42,752,822 cards   per-card obs/exp range 0.9732 (Ah) .. 1.0188 (2h)
          flop suits: rainbow 39.74% (exp 39.76%)  two-tone 55.07% (exp 55.06%)  monotone 5.19% (exp 5.18%)
          flop ranks: unpaired 82.79% (exp 82.82%)  paired 16.97% (exp 16.94%)  trips 0.24% (exp 0.24%)
```

Strongly indicates our dataset is fair. Top 6 are different HandHQ venues, overall HandHQ at the bottom (what we kept). Nearly on the expected for every category. Worth noting: a pattern that sticks out here is that aces appear to be underrepresented while 2s/3s are overrepresented. This selection bias happens due to the flop cards only being recorded when the flop is reached, which typically happens when better cards are in play, especially aces.

Hole Data:

Legend

- 72o: 7-2 offsuit
- AAs: A-A suited

Pluribus isolated:

```
pluribus (all seats revealed: unbiased control): 60,000 known hole card pairs
  pairs 5.69% (exp 5.88%)   suited 23.64% (exp 23.53%)   offsuit 70.67% (exp 70.59%)
  most over-represented:  43s 1.23x  72s 1.20x  73s 1.17x  96s 1.16x  QTs 1.15x  J4o 1.13x  J8s 1.13x  A7s 1.12x
  most under-represented: AJs 0.90x  A3s 0.90x  86s 0.90x  KQs 0.89x  54s 0.89x  J2s 0.85x  64s 0.85x  T9s 0.77x
  obs/exp ratio grid (rows/cols A..2; upper-right suited, lower-left offsuit):
       A    K    Q    J    T    9    8    7    6    5    4    3    2
    A 0.94 0.99 1.03 0.90 1.00 1.06 1.03 1.12 0.90 0.96 0.99 0.90 1.01
    K 0.97 0.96 0.89 1.12 1.02 0.95 1.05 1.04 0.94 0.91 1.04 1.10 1.03
    Q 0.93 0.94 0.90 1.02 1.15 1.02 0.99 1.02 1.10 0.94 1.01 1.06 0.93
    J 0.99 0.98 0.96 0.95 1.04 1.03 1.13 0.99 1.06 0.98 1.00 0.99 0.85
    T 1.02 1.00 0.98 1.04 0.94 0.77 0.92 1.07 0.97 0.97 0.95 1.03 0.98
    9 1.02 0.96 0.99 0.97 0.94 1.04 0.98 1.03 1.16 0.97 1.10 0.97 0.97
    8 1.06 1.02 1.00 0.97 0.96 1.00 0.92 0.92 0.90 0.91 0.92 1.06 0.91
    7 0.99 1.00 0.99 0.97 1.09 0.97 1.06 0.95 1.10 1.04 1.00 1.17 1.20
    6 1.09 0.92 1.00 1.09 0.97 1.04 1.12 0.92 0.94 1.10 0.85 0.93 1.06
    5 1.08 1.02 1.01 1.02 0.96 1.01 1.02 1.06 1.04 0.98 0.89 1.04 1.04
    4 1.00 1.04 0.99 1.13 0.94 0.98 0.99 0.94 1.06 1.01 1.01 1.23 1.09
    3 0.94 0.99 0.98 0.90 1.00 1.01 1.01 0.99 0.96 1.04 0.94 1.05 0.93
    2 0.94 1.04 0.99 0.99 1.07 1.06 1.06 1.00 1.01 0.98 0.99 0.97 1.01
```

Decent distribution, but these came from Pluribus bot games, which are now removed.

All Known HandHQs:

```
HandHQ (shown-down cards only: selection biased): 5,625,693 known hole card pairs
  pairs 21.82% (exp 5.88%)   suited 30.72% (exp 23.53%)   offsuit 47.46% (exp 70.59%)
  most over-represented:  AA 5.50x  KK 5.41x  QQ 5.20x  JJ 4.80x  TT 4.37x  AKs 4.27x  AKo 4.03x  99 3.95x
  most under-represented: T2o 0.20x  73o 0.19x  93o 0.18x  83o 0.18x  62o 0.17x  72o 0.17x  92o 0.17x  82o 0.17x
  obs/exp ratio grid (rows/cols A..2; upper-right suited, lower-left offsuit):
       A    K    Q    J    T    9    8    7    6    5    4    3    2
    A 5.50 4.27 3.76 3.34 2.97 2.44 2.26 2.15 2.00 2.09 1.98 1.92 1.84
    K 4.03 5.41 3.06 2.70 2.36 1.71 1.37 1.23 1.10 1.03 0.97 0.91 0.88
    Q 3.34 2.41 5.20 2.62 2.26 1.66 1.26 0.98 0.84 0.79 0.73 0.69 0.64
    J 2.79 1.96 1.75 4.80 2.44 1.80 1.36 1.00 0.73 0.67 0.60 0.56 0.54
    T 2.19 1.49 1.36 1.42 4.37 2.12 1.52 1.13 0.76 0.58 0.51 0.47 0.44
    9 1.45 0.86 0.77 0.83 0.94 3.95 1.92 1.41 0.94 0.64 0.48 0.42 0.39
    8 1.23 0.61 0.53 0.56 0.56 0.76 3.59 1.77 1.23 0.83 0.56 0.41 0.36
    7 1.09 0.52 0.40 0.38 0.39 0.47 0.65 3.23 1.62 1.08 0.70 0.47 0.38
    6 0.96 0.45 0.34 0.29 0.28 0.33 0.39 0.56 2.89 1.48 0.95 0.62 0.43
    5 0.98 0.41 0.30 0.27 0.24 0.24 0.27 0.34 0.48 2.65 1.27 0.78 0.54
    4 0.90 0.37 0.28 0.24 0.21 0.20 0.21 0.24 0.29 0.40 2.41 0.94 0.57
    3 0.82 0.35 0.26 0.23 0.20 0.18 0.18 0.19 0.22 0.26 0.29 2.21 0.65
    2 0.76 0.33 0.25 0.22 0.20 0.17 0.17 0.17 0.17 0.20 0.20 0.22 2.02
```

Notable caveat on HandHQ, the dataset was sampled in a way that unrevealed cards couldn't be tracked (represented with ???? in the original dataset). So, if a game ended with 4 players still in and the first player to reveal had the best hand, leading to the remainder of the board to not reveal their hands, they were never recorded. Only ~5% of the dataset has known hole card pairs.

We should not remove these games as this would lead to heavily biased datasets that would behave as if AA or KK are significantly more likely to be in play than they are.

High level inventory:

General:

```
Variants seen: {'NT': 21616099, 'FO/8': 14, 'F7S': 13, 'FR': 10, 'PO': 8, 'F2L3D': 8, 'FT': 7, 'N2L1D': 7, 'F7S/8': 7, 'FB': 1, 'NS': 1}
bot games (excluded): 10,000   Kept NLHE human hands: 21,606,099
Duplicate hand ids among kept hands: 400
```

Some bot games, non-No Limit Texas Hold'em games, and duplicates were removed.

Player distributions:

```
 2 players:    3,054,696   14.1%
 3 players:      657,764    3.0%
 4 players:    1,991,987    9.2%
 5 players:    5,148,335   23.8%
 6 players:    6,135,744   28.4%
 7 players:      873,272    4.0%
 8 players:    1,782,538    8.3%
 9 players:    1,735,586    8.0%
10 players:      226,177    1.0%
```

Did not use this for any justification today, may be useful for extensions of the initial model.

Street reached (stage of the game played to):

```
preflop   11,211,640   51.9%
flop       3,736,537   17.3%
turn       1,746,367    8.1%
river      4,911,555   22.7%
```

Most games end at the preflop, making this the most important stage for us to hone in on for our initial model of predicting preflop behavior.

Big blind sizes (top 10, vast majority <10):

```
  0.25:    5,966,811
     4:    3,855,571
     2:    3,292,466
     1:    2,356,096
    10:    2,199,889
   0.5:    2,139,677
     6:    1,792,784
   0.1:        2,004
     3:          649
     5:           64
```

Supporting justification on removing wsop data. Big blinds in those games were several tens or hundreds of thousands, significant outliers.

Removed hands:

source	source	bot	non-nlhe	duplicate	kept
handhq	0	0	0	15	21,606,072
misc	5	0	0	0	0
pluribus	0	10,000	0	0	0
wsop	83	0	0	0	0

- Removed the misc and wsop sources
    - misc served as templates/examples
    - wsop was mostly not No Limit Texas Hold'em, only about 20 games were left. Decided to remove as it was a small subset of games with significant outliers in the chip data

Data Transformation:

```
[1]
variant = 'NT'
ante_trimming_status = false
antes = [2.50, 2.50, 2.50, 2.50, 2.50, 2.50]
blinds_or_straddles = [5, 10, 0, 0, 0, 0]
min_bet = 10
starting_stacks = [2067.40, 1987.50, 1171, 1710.50, 2000, 1515]
actions = ['d dh p1 ????', 'd dh p2 ????', 'd dh p3 ????', 'd dh p4 ????', 'd dh p5 ????', 'd dh p6 ????', 'p3 cbr 47.50', 'p4 f', 'p5 f', 'p6 f', 'p1 f', 'p2 f']
venue = 'Absolute Poker'
time = 00:00:26
day = 1
month = 7
year = 2009
hand = 3017243630
seats = [3, 4, 5, 6, 1, 2]
table = 'CLEO AVE'
players = ['X+u4T/E5ANkyZLKm1YjqwQ', 'wyXD1O26Buq3VWHAij37Jg', '3wT3m+GDGtVWU1KR2MWJ1Q', '/P+7Z0P/b7YiK60FW9dRAQ', 'eXXdS46B0E4apgZgp7gHFw', 'XrM1XlN29RxmLx3oZHhG0w']
winnings = [0, 0, 37.50, 0, 0, 0]
currency_symbol = '$'
time_zone_abbreviation = 'ET'
```

Original PHH format:

1. variant: Variant of poker (NT = No Limit Texas Hold'em)
2. ante_trimming_status: ante trimming indicator flag
3. antes: Per-player antes
4. blinds_or_straddles: Per-player blinds/straddles
5. min_bet: Minimum bets
6. starting_stacks: Per-player starting currency
7. actions: Game sequence split per event
    a. All players were dealt cards (dh)
    b. P3 bet 47.50 (cbr)
    c. All other players folded (f)
8. venue: Game host
9. time: HH:MM:SS time
10. day: Day
11. month: Month
12. year: Year
13. hand: Unique hand ID (per venue)
14. seats: Physical seat location of the players
15. table: Table name, more relevant for games at physical venues, obfuscated for online play
16. players: Player uIDs
17. winnings: Earnings summary from the game
18. currency_symbol: Currency indicator
19. time_zone_abbreviation: Time zone

New parquet data format:

antes	blinds_or_straddles	min_bet	starting_stacks	actions	winnings
Missing: 0 (0%) Distinct: 8 (<1%)	Missing: 0 (0%) Distinct: 19 (<1%)	Missing: 0 (0%) Distinct: 2 (<1%)	Missing: 0 (0%) Distinct: 97624 (98%)	Missing: 0 (0%) Distinct: 69570 (70%)	Missing: 21287 (21%) Distinct: 4718 (5%)
[0. 0. 0. 0. 0. 0.] 25% [0. 0. 0. 0. 0.] 16% [0. 0.] 16% Other 42%	[0.25 0.5 0. 0. 0. 0.] 25% [0.25 0.5] 16% [0.25 0.5 0. 0. 0.] 16% Other 42%	Min 0.25 Max 0.5	97624 Distinct values	69570 Distinct values	4718 Distinct values
[0. 0. 0. 0. 0.]	[0.25 0.5 0. 0. 0.]	0.5	[12.82 147.69 17.9 46.04 55.3]	['d dh p1 ????' 'd dh p2 ????' 'd dh p.	[0. 0.75 0. 0. 0.]
[0. 0.]	[0.25 0.5]	0.5	[89.54 14.07]	['d dh p1 ????' 'd dh p2 ????' 'p2 f']	[0.75 0.]
[0. 0. 0. 0. 0. 0.]	[0.25 0.5 0. 0. 0. 0.]	0.5	[41.2 25.15 24.2 46.25 38.84 173.52]	['d dh p1 ????' 'd dh p2 ????' 'd dh p.	[0. 0. 0. 1.25 0. 0.]
[0. 0.]	[0.25 0.5]	0.5	[13.82 89.79]	['d dh p1 ????' 'd dh p2 ????' 'p2 cc' '	[0. 0.95]
[0. 0.]	[0.25 0.5]	0.5	[90.24 13.32]	['d dh p1 ????' 'd dh p2 ????' 'p2 cbr	[0. 1.]
[0. 0. 0. 0. 0. 0.]	[0.25 0.5 0. 0. 0. 0.]	0.5	[49.5 70.69 12. 18.5 26.15 117.05]	['d dh p1 ????' 'd dh p2 ????' 'd dh p.	[0. 0. 0. 4.55 0. 0.]
[0. 0. 0. 0. 0.]	[0.25 0.5 0. 0. 0.]	0.5	[18.85 46.25 43.25 52.83 16.25]	['d dh p1 ????' 'd dh p2 ????' 'd dh p.	[0. 0. 0. 3.35 0.]
[0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]	[0.25 0.5 0. 0. 0. 0. 0. 0. 0. 0.]	0.5	[13.95 32. 11.25 61.69 25.3 37.4 24.3	['d dh p1 ????' 'd dh p2 ????' 'd dh p.	[1. 0. 0. 0. 0. 0. 0. 0. 0.]
[0. 0.]	[0.25 0.5]	0.5	[13.82 89.74]	['d dh p1 ????' 'd dh p2 ????' 'p2 cc' '	[1. 0.]
[0. 0. 0. 0. 0.]	[0.25 0.5 0. 0. 0.]	0.5	[147.94 17.9 46.04 55.3 12.57]	['d dh p1 ????' 'd dh p2 ????' 'd dh p.	[0. 0. 1.25 0. 0.]
[0. 0. 0. 0. 0. 0. 0. 0. 0.]	[0.25 0.5 0. 0. 0. 0. 0. 0. 0.]	0.5	[48.72 41.98 16.6 21.7 74.87 10.4 18.(	['d dh p1 ????' 'd dh p2 ????' 'd dh p.	[0. 0. 0. 0. 0. 0. 0. 0. 9.5]
[0. 0.]	[0.25 0.5]	0.5	[89.24 34.32]	['d dh p1 ????' 'd dh p2 ????' 'p2 cbr	[0. 8.55]
[0. 0. 0. 0. 0. 0. 0. 0.]	[0.25 0.5 0. 0. 0. 0. 0. 0.]	0.5	[129.78 95.19 45.55 20.7 45.75 50.83	['d dh p1 ????' 'd dh p2 ????' 'd dh p.	*Missing value*
[0. 0. 0. 0. 0. 0. 0. 0.]	[0.25 0.5 0. 0. 0. 0. 0. 0.]	0.5	[31.5 61.69 25.3 37.4 24.3 36.39 32.0(	['d dh p1 ????' 'd dh p2 ????' 'd dh p.	[0. 0. 0. 0. 0. 0. 3.6 0.]
[0. 0. 0. 0. 0.]	[0.25 0.5 0. 0. 0.]	0.5	[24.65 47. 38.84 173.52 40.95]	['d dh p1 ????' 'd dh p2 ????' 'd dh p.	*Missing value*
[0. 0. 0. 0. 0.]	[0.25 0.5 0. 0. 0.]	0.5	[17.4 55.3 20. 12.57 147.69]	['d dh p1 ????' 'd dh p2 ????' 'd dh p.	[0. 0. 5.95 0. 0.]

`Hands` Parquet Table Snippet Visualized with Microsoft's Data Wrangler

The PHH entries were copied to Parquet tables to serve as lightweight typed databases that can be ingested with Pandas much faster than parsing the plaintext PHH formats as is. This yielded ~10x speedup for the metrics parsing scripts from a local device, indicating this format is more scalable for larger datasets and will allow us to train and test quicker.

The PHH formats have been retained in the Github repository and may still be used in some capacity since it is much easier for everyone to review diffs to the PHH representation, the parquet tables can be easily rebuilt, it is the intended format for use with the pokerkit library, and we have room for it since the plaintext can be compressed.

The dataset was split into 2 tables: hands and decisions. The former represents nearly a direct transformation of the PHH format that drops some irrelevant data (players, venue, time, table, seats, finishing stacks) where one row carries the entire game's data. The latter displays a sequential replay of the game where every row shows a player action alongside the current state of the game.

The decisions table will serve as the primary training data for predicting player behavior during the pre-flop. We are keeping more data than we need from the games in case we are able to expand the functionality of our model.